



IF3110 – Web-based Application Development

Introduction

Objectives

- › Understanding on how Internet works, how Internet-based Application works
- › Ability to develop an application using various technologies at server & client side
- › Ability to design an Internet application w/ considering various qualities (e.g., security, performance, usability)

Course Activity

- › Courses: 3 credits

- › 3 hours lecture – Monday 07.00 – 08.40 and Wednesday (09.00 – 10.00)

- › 6 hours independent activities (e.g., exploration, assignment)

- › Lecturers:

- › K1: Yudistira Asnar (yudis@itb.ac.id) & Bayu Samudra (bayu.samudra@itb.ac.id)

- › K2: Mohammad Koyyim (koyimatu@itb.ac.id) & Muhammad Zuhri Catur Candra (mzc@itb.ac.id)

- › K3: Riza Satria Perdana (riza@itb.ac.id) & Galuh Dipa Bharata (galuhdb@itb.ac.id)

- › Assessment

- › Assignments: 1 (continuous project)

- › Exams: Quiz, Mid, and Final

- › Note: Assignments are dominant (~55:~45)

- › This might be slightly adjusted based on the participants' performance

Course Activity: Assignment

- › It is done in a group of 3 students (defined by lecturers)
- › It must be submitted following the procedure defined by assistants
- › Some assignments need to be presented to the assistants
- › More details will be defined at the assignment's Term of References

Noteworthy

- › All forms of cheating will be punished severely
 - › Faking your attendance
 - › Copying others' works
 - › Ask LLM, such as GPT/Claude (or something similar), to write code for your project

Topics

- › Basic Concepts of Internet and Internet Application
- › HTTP
- › HTML + Java Script + CSS
- › Client Side: JS and its advances
- › Server Side
- › Data Access
- › AJAX, Single Page Application
- › Web Framework: Node.js++
- › Cloud & Cloud Native
- › Web Service
- › GraphQL, APIM
- › Event-Based Web & Asynchronous Processing
- › Web Security
- › Intro to
 - › Performance and Security Engineering
 - › Web Testing
 - › Usability
 - › Microservice

References (some)

- Full Stack Web Development: The Comprehensive Guide, Philip Ackermann, Rheinwerk Computing, 2023
- Web Technologies: A Computer Science Perspective, Jeffrey C. Jackson, Prentice Hall, 2006
- Developing Large Web Applications: Producing Code That Can Grow and Thrive, Kyle Loudon, O'Reilly Media, Inc., 2010
- HTML5: Up and Running, Mark Pilgrim, O'Reilly Media, Inc., 2010
- JavaScript: The Definitive Guide, David Flanagan, O'Reilly Media, 2011
- **Remark:** slides and books are just the guideline to follow the course. Use Internet for complete references

Communication

- › Synchronous will held at the Class
- › Edunex <https://edunex.itb.ac.id/> (Parent Class)
- › Microsoft Teams (joint code **cg1y8rj**)
 - › Member: lecturers, assistants, students
 - › Purpose :
 - › announcement (esp. lecturer/assistant → students)
 - › discussion relevant to the course
- › Lecturers' email
 - › yudis@itb.ac.id
 - › koyimatu@itb.ac.id
 - › mzc@itb.ac.id
 - › riza@itb.ac.id
 - › with subject: [IF3110] xxxx
- › Remark: This means can be modified as the best suited for the lecturers

Any additional topic requests?

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